

Lab 1 — First Commit • Tonight in 20 Steps

OPIM 5512 • Module 1 — Hartford Sep 2, Stamford Sep 9 (same lab, run twice)

The one-sentence version: tonight the deliverable is a *repo*, not a notebook. Two people each analyze a different half of the same problem, and the only way to the answer is to merge your work together.

Before we start (stack check)

- [] **GitHub account + GitHub Desktop** installed and **signed in**
 - [] **Google Colab** opens
 - [] You know your **campus station**: Hartford = `KBDL`, Stamford = `KBDR`
 - [] **No AI tonight** — no Claude / ChatGPT / Copilot / Colab autocomplete. Type it yourself. (Module 3 is an entire unit on using them. Not never — not yet.)
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The 20 steps, in order

Set up (steps 1–6) 1. **Pair up.** Decide who's **Partner A (weather)** / **Partner B (demand)**. (*Solo/online? Two accounts, or pair over Teams.*) 2. **Partner A:** create a new **public** repo for the pair. 3. **Partner A:** add Partner B as a **collaborator** → **Partner B accepts** the invite. 4. **Partner A:** turn on **branch protection** for `main` (require a PR + 1 approval). 5. **Both:** **clone** the repo **once** in GitHub Desktop. 6. **Each:** make your branch (`dev-weather` / `dev-demand`) → **Publish** it.

Grab data & build your half (steps 7–13) 7. **Each:** open your **starter** in Colab (new tab) → **run the first cell, look at your raw data**. 8. **Together:** write the **data dictionary** (your contract) in `README.md` — names, units, what one timestamp means. 9. **Each:** clean your half in Colab → **File** → **Save** to your branch (*drop a `#TEST` to prove it saved*). 10. **Each:** save your **CSV** and your **plot**; download both. 11. **Each:** GitHub Desktop → **Show in Explorer** → drag CSV → `data/clean/`, plot → `images/`. 12. **Each:** **commit** (real message) → **push**. 13. **Each:** open **one pull request** (your branch → `main`): notebook + contract + CSV + plot.

Review & merge (steps 14–17) 14. **Each:** request your **partner** as reviewer. 15. **Each:** **review your partner's PR** — read the diff, then **Approve**. 16. **Merge** both PRs → **delete** the branches. (*Whoever merges the README second resolves the merge conflict — keep both halves.*) 17. **Both:** switch to `main` → **Fetch** → **Pull**. Now you each have **both** files.

Join & report (steps 18–20) 18. **Together (one screen):** open the **joint EDA** notebook → run the **join** → find **three things**, one surprising. 19. **Write** `REPORT.md` (from the template) + save **figures** → branch → PR → partner reviews → merge. 20. **Read-out:** one figure on screen, one sentence — and check your **network graph** shows the loop both ways.

Don't panic

- **Commit early.** Once something's committed, it's essentially impossible to lose.
- **Fetch and Pull can't hurt you.** Fetch = peek, Pull = download. Neither overwrites your work. The scary buttons are the *Discard* ones.

- **Save to the same path every time** — a different path makes a *second* notebook, not an update.
 - **Restart & run all** → **Clear all outputs** before a notebook goes in — your partner has to read that diff.
 - `main` **rejecting your push is the protection working, not a bug** — you're on `main` when you should be on your branch.
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Online / solo students

- **Pair over Teams** if two of you can — each on your own account, one owns the repo and adds the other. Real review, real contract.
 - **Otherwise two accounts** (needs a second email): play both A and B yourself — you'll hit every step, including approving as the other account and resolving the conflict.
 - **Simplest solo:** one account with **Required approvals = 0** — branch → PR → merge yourself (skips the approval gate).
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Definition of done — what's in the repo at 7:30

- [] Both partners are collaborators; **branch protection** on `main`
- [] `README.md` with a data dictionary for **both** files
- [] `data/clean/weather_hourly.csv` **and** `data/clean/demand_hourly.csv`
- [] Both cleaning notebooks + the joint EDA notebook
- [] `REPORT.md` — three findings (one surprising), figures, honest caveats, plain English
- [] **≥3 merged pull requests**, branches deleted, both of you authoring **and** reviewing
- [] A **network graph** showing the loop going both ways